

ALEKSANDR "ALEX" KRASNOK

Ph.D. | Assistant Professor, Electrical & Computer Engineering
Florida International University | Director, QTM Lab

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200+

scholarly works

15,222

citations

61

h-index

45+

journal papers at FIU

Google Scholar, September 25, 2026; FIU publication count since August 2021.

Research in complex-frequency electrodynamics, quantum control, nanophotonics, and sensing. Develops analytical models and computational methods in collaboration with experimental teams working on optical, microwave, and cryogenic systems.

APPOINTMENTS & LEADERSHIP

Assistant Professor; Director, Quantum Technologies & Metamaterials Lab

Florida International University (FIU)

Aug 2021 - Present

- Established the QTM Lab and authored or coauthored **45+ journal publications**, including work in *Science* and *Physical Review Letters*. Courtesy appointment in FIU's Knight Foundation School of Computing and Information Sciences.
- Co-founder, FIU Quantum Initiative** and core member of its university-wide committee. Contributes quantum-hardware, photonics, and education expertise to planning for FIU's announced **IonQ Superion 256** system, expected on campus in late 2027.
- Collaborates with CUNY, RadiaBeam, Chicago/Argonne, Calgary, Simon Fraser, and LAPLACE-Toulouse on quantum circuits, superconducting resonators, and plasma experiments.

Research Assistant Professor; Founding Director, Photonics Core Facility

CUNY Advanced Science Research Center, New York

Jul 2018 - Jul 2021

- Built shared-user operations from the ground up: approximately **eight optical tables** and **100+ academic and industry users** by 2021. Developed laser, spectroscopy, and infrared ellipsometry capabilities.
- 30+ journal publications** during the appointment. Contributed theory to the 2020 *Nature* photonic-twistronics study, a *Physics World* Top 10 Breakthrough of 2020.

Postdoctoral Researcher / Research Scientist

The University of Texas at Austin

Nov 2016 - Jul 2018

- 20+ journal publications** in Andrea Alù's group. Worked with experimental teams on quantum-dot photonics, exciton-plasmon coupling, and coherent wireless power transfer.

Senior Research Fellow, Photonics / Electrodynamics

ITMO University, Saint Petersburg

Jun 2013 - Nov 2016

- Led an all-dielectric nanophotonics team of **two postdoctoral fellows and approximately five students**; authored or coauthored **20+ journal publications** during the appointment.

EDUCATION

Ph.D., Photonics and Quantum Optics - ITMO University, 2013.

M.S. / B.S., Quantum Optics - Far Eastern Federal University, 2010 / 2008.

SELECTED HONORS

2026 - Rising Star in Engineering, Academy of Science, Engineering and Medicine of Florida.
2026-2029 - Selected, Fulbright Specialist Roster, U.S. Department of State.
2026 - Finalist, FIU Presidential Excellence Award. **2024-2025** - FIU Top Scholar (Research).
2024 - Leopold B. Felsen Award for Excellence in Electrodynamics, EurAAP; IEEE Senior Member.

SELECTED RESEARCH SUPPORT

DOE / Brookhaven National Laboratory, EI-Pho - FIU subcontract PI; **\$600,000 ceiling**, with **\$60,000 initially allotted** and incremental funding; base period 2026, options through 2029.
AFOSR - PI, complex-frequency plasma research; **\$50,000**, 2026.
NSF I-Corps, Meta-MRI - PI; **\$50,000**, 2024-2026.
FIU start-up - **\$500,000**, plus separate support for three graduate students over two academic years.

SELECTED PUBLICATIONS

A. Krasnok. **Constant-Amplitude 2π Phase Modulation from Topological Pole-Zero Winding**. *Physical Review Letters* 137, 053801 (2026).
D. Trivedi, A. Bhowmik, L. Niaz, A. Krasnok. **Reflection-Zero Waveforms for Selective Loading in Shared-Line Superconducting Qubit Circuits**. *Advanced Quantum Technologies* 9, e70441 (2026). Selected for a cover feature.
S. Kim, A. Krasnok, A. Alù. **Complex-Frequency Excitations in Photonics and Wave Physics**. *Science* 387, eado4128 (2025).
A. Krasnok et al. **Superconducting Microwave Cavities and Qubits for Quantum Information Systems**. *Applied Physics Reviews* 11, 011302 (2024).
G. Hu, Q. Ou, G. Si, Y. Wu, J. Wu, Z. Dai, A. Krasnok, et al. **Topological Polaritons and Photonic Magic Angles in Twisted α -MoO₃ Bilayers**. *Nature* 582, 209-213 (2020).

Complete publication record: Google Scholar

TEACHING & MENTORSHIP

Developed FIU's undergraduate and graduate quantum-computing courses (EEE 4423 / EEE 6429), integrating Qiskit, Python, and hardware exercises. Published **70+ recorded teaching modules**; both courses received the *Affordability Counts* medallion.

Author, *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide* (Springer, forthcoming 2026).

At FIU, supervised **three postdoctoral fellows**, co-supervised **one Ph.D. to completion**, and mentored **11 undergraduate researchers and Senior Design students**. Supports technical independence and career development; former trainees now work at imec USA Florida and Nokia.

PROFESSIONAL SERVICE

Associate Editor: *Photonics Research* (2022-present). **Editorial boards:** *Advanced Photonics Nexus* (2024-present) and *Reviews on Advanced Materials Science* (2023-present). Organizer, FIU Qiskit Fall Fest (2025); proposal reviewer for DOE and NSF.